

Question #1 of 40

Question ID: 1474068

Charles Griffith makes quarterly bets between stocks of industrial and utility sectors. The historical correlation between the returns of the two sectors is -0.20 and Griffith's bets have been correct 55% of the time. Further information is as below:

Sector	E (R)	σ	Benchmark
			Weight
Industrial	12.00%	13.0%	80%
Utility	5.2%	2.5%	20%

The expected annualized active return of Griffith's sector rotation strategy is *closest* to:

- A) 13.72%
 - B) 10.64%
 - C) 5.48%
-

Question #2 of 40

Question ID: 1474066

Charles Griffith makes quarterly bets between stocks of industrial and utility sectors. The historical correlation between the returns of the two sectors is -0.20. Further information is as below:

Sector	E (R)	σ	Benchmark
			Weight
Industrial	12.00%	13.0%	80%
Utility	5.2%	2.5%	20%

The annualized active risk of Griffith's strategy is *closest* to:

- A) 13.72%
- B) 10.90%
- C) 27.44%

Question #3 of 40

Question ID: 1474052

Which of the following is correct for a constrained active portfolio?

- A) $TC=1$
- B) $TC<1$
- C) $TC>1$

Radina Radichkova, CFA, is considering investing in one of three actively managed funds whose benchmark is the FTSE 100. The Sharpe ratio and standard deviation of the benchmark are 0.50 and 15%, respectively.

	Alpha	Bankso	Crystal
Active return	3.2%	2.8%	12.5%
Active risk	4.1%	3.6%	15.5%

Radichkova is also analyzing an actively managed portfolio consisting of financials and pharmaceuticals. The following table shows the weights and returns of the benchmark and portfolio:

Sector	WB	WP	RB	RP
Financials	50%	70%	9.2%	17.8%
Pharmaceuticals	50%	30%	8.1%	6.3%

Radichkova makes the following comments about the two-sector portfolio:

Comment 1: The value added is 5.7%.

Comment 2: The asset allocation decision only accounts for 0.22% of the value added.

Radichkova is writing a summary of the fundamental law of active management. In that section of the report, she states that the transfer coefficient can be viewed as the correlation

between:

1. ex-ante active returns and actual active weights
2. risk-weighted optimal active weights and risk-weighted actual active weights
3. ex-ante active returns and realized active returns

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Question ID: 1474059

In relation to funds Alpha, Bankso, and Crystal, the highest achievable Sharpe ratio is *closest* to:

- A) 0.85.
 - B) 0.90.
 - C) 0.95.
-

Question #5 - 7 of 40

Question ID: 1474060

If the FTSE 100 and Crystal fund are combined in an optimal portfolio, what proportion should be invested in Crystal?

- A) 136%.
 - B) 146%.
 - C) 156%.
-

Question #6 - 7 of 40

Question ID: 1474061

How many of Radichkova's comments are correct in relation to the two-sector portfolio?

- A) One.
 - B) Both.
 - C) None.
-

Question #7 - 7 of 40

Question ID: 1474062

Which are the correct definitions of the transfer coefficient included in Radichkova's report?

- A) 1 and 2.
- B) 2 and 3.
- C) 1 and 3.

Question #8 of 40

Question ID: 1474055

An active manager currently covers 40 stocks and makes a forecast for each of them every quarter. Next year he intends to cover the same stocks but only once every 6 months. Assuming the manager's skill, measured in terms of the correlation of each forecast with actual returns doesn't change, which of the following statements is *most* accurate?

- A) The information ratio will fall by approximately 30%
- B) The information ratio will fall by approximately 50%
- C) The information coefficient will fall by approximately 50%

Question #9 of 40

Question ID: 1474034

Helen Wilde is trying to estimate the active return of Optimal fund. A comparison of Optimal's holdings and that of the benchmark are shown below:

Asset Class (i)	Optimal Weight (w_{Pi})	Benchmark Weight (w_{Bi})	Optimal Return $E(R_{Pi})$	Benchmark Return $E(R_{Bi})$
Industrials	30%	40%	11%	12%
Financials	50%	30%	6%	5%
Utilities	20%	30%	14%	12%

The expected active return due to asset allocation for Optimal is *closest* to:

- A) – 1.40%

B) – 0.86%

C) -0.44%

Question #10 of 40

Question ID: 1474040

Which of the following statements is *least accurate*?

A) A closet index fund has a low Sharpe ratio.

B) The Sharpe ratio of a portfolio is unaffected by addition of cash or leverage in the portfolio.

C) Investors can take active risk that is suitable for them by investing in a combination of actively managed portfolio and benchmark portfolio.

Question #11 of 40

Question ID: 1474054

An active manager expects his information coefficient to drop from 0.08 to 0.02 in the coming period due to extremely volatile and unpredictable markets. As a response he intends to increase his breadth by a factor of 4. Which of the following statements is *most accurately* describes the impact on the information ratio?

A) The information ratio will increase

B) The information ratio will decrease

C) The information ratio will remain constant

Question #12 of 40

Question ID: 1474064

Jon Gamlin is comparing a market timing strategy with a stock selection strategy. He draws the following two conclusions for unconstrained active managers:

Conclusion 1

To achieve the same information ratio, a market timer making weekly forecasts on the movement of the market needs to have a higher skill level than a stock selector following 25 stocks and updating the forecast semi-annually

Conclusion 2

A specialist following only 4 stocks who revises his forecast 100 times per year will achieve the same information ratio as a stock selector with the same skill level who follows 50 stocks and updates his assessments semi-annually

Regarding Gamlin's conclusions:

- A)** Only conclusion 2 is correct.
 - B)** Only conclusion 1 is correct.
 - C)** Neither conclusion is correct.
-

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Question ID: 1474038

Zeta fund has active return and active risk of 1.6% and 8% respectively. Benchmark portfolio has a Sharpe ratio of 0.35 and standard deviation of benchmark returns is 10.5%.

What is the level of active risk that an investor would need to take to maximize the Sharpe ratio of a portfolio consisting of Zeta fund and the benchmark portfolio?

- A)** 8%
 - B)** 7%
 - C)** 6%
-

Question #14 of 40

Question ID: 1474074

An active manager makes quarterly bets on the stocks in the Russell 2000 index and uses the index as the benchmark. The manager claims a modest IC of 0.02 using a stock screening model. Sam Fox, CFA makes the following two statements:

- I. The bets on the 2000 stocks in the index is not independent as the screens by definition introduce dependency in the decision process.
- II. The quarterly bets are likely to be independent.

How many of Fox's statements are correct:

- A)** Both statements are correct.
 - B)** Neither statement is correct.
 - C)** Only one statement is correct.
-

Question #15 of 40

Question ID: 1474035

Helen Wilde is trying to estimate the active return of Optimal fund. A comparison of Optimal's holdings and that of the benchmark are shown below:

Asset Class (i)	Optimal Weight (w_{Pi})	Benchmark Weight (w_{Bi})	Optimal Return $E(R_{Pi})$	Benchmark Return $E(R_{Bi})$
Industrials	30%	40%	11%	12%
Financials	50%	30%	6%	5%
Utilities	20%	30%	14%	12%

The expected active return for Optimal is *closest* to:

- A)** -0.44%
 - B)** - 0.80%
 - C)** - 1.40%
-

Question #16 of 40

Question ID: 1474057

Alisa Darent is evaluating several active portfolio managers with the same style and benchmark portfolio.

Manager	Active return	Active risk
Alfred	3.00%	12.00%
Brad	2.20%	11.00%
Charles	2.00%	10.50%

Benchmark return is expected to be 11%. What will be the maximum expected return for Darent's portfolio assuming that she wants to limit her active risk to 11%?

- A) 13.75%
- B) 2.75%
- C) 2.20%

Question #17 of 40

Question ID: 1474037

Zeta fund has active return and active risk of 1.6% and 8% respectively. Benchmark portfolio has a Sharpe ratio of 0.35 and standard deviation of benchmark returns is 10.5%.

The maximum possible Sharpe ratio of a portfolio consisting of Zeta fund and the benchmark portfolio is *closest* to:

- A) 0.55
- B) 0.4
- C) 0.5

Ufton Wealth Management's Ranger fund has proved popular with clients. An extract from the prospectus of the Ranger fund is shown in Exhibit 1.

Exhibit 1: Ranger Fund

Asset	Portfolio weight	Benchmark weight	Expected portfolio return	Expected benchmark return
U.S. equities	15%	20%	11%	9%

U.S. corporate bonds	35%	35%	8%	7%
International equities	8%	40%	14%	10%
U.S. real estate	42%	5%	7%	7%

Ufton awards its best performing fund manager with a large cash bonus each year. Details of the performance of three funds is shown in Exhibit 2. Risk-free rate is 2%.

Exhibit 2: Selected Fund Performance

Fund	Portfolio return	Benchmark return	Portfolio standard deviation	Benchmark standard deviation	Sharpe ratio	Tracking error
Saltire	8.46%	5.80%	6.13%	4.50%	1.05	1.58%
Dragon	13.01%	11.56%	7.64%	5.15%	1.44	2.12%
Rose	11.39%	11.37%	11.01%	11.14%	0.85	0.21%

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Question ID: 1474042

The expected level of active return expected to be achieved by the Ranger fund is *closest* to:

- A) -9.00%.
- B) -0.09%.
- C) +3.49%.

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Question ID: 1474043

The largest positive contribution to the active return achieved by the Ranger fund is expected to come from:

- A)** security selection.
 - B)** asset allocation.
 - C)** Cannot tell from the information available.
-

Question #20 - 21 of 40

Question ID: 1474044

Of the three funds described in Exhibit 2, the fund with the highest information ratio is:

- A)** Saltire.
 - B)** Dragon.
 - C)** Rose.
-

Question #21 - 21 of 40

Question ID: 1474045

Of the three funds described in Exhibit 2, the *most* likely to be a closet index fund is:

- A)** Saltire.
 - B)** Dragon.
 - C)** Rose.
-

Question #22 of 40

Question ID: 1474046

Which of the following terms is the cross-sectional correlation between forecasted active returns and actual active weights adjusted for risk?

- A)** Transfer Coefficient
 - B)** Breadth
 - C)** Information Coefficient
-

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Question ID: 1474036

Zeta fund has active return and active risk of 1.6% and 8% respectively. Benchmark portfolio has a Sharpe ratio of 0.35 and standard deviation of benchmark returns is 10.5%.

What is the weight of benchmark portfolio in a portfolio consisting of Zeta fund and the benchmark portfolio assuming that the portfolio is constructed to have optimal active risk?

- A) 0.1667
 - B) 0.2
 - C) 0.25
-

Question #24 of 40

Question ID: 1474033

Susan Thomas is evaluating the holdings of Primus fund. Based on the information below, the estimated active return is *closest* to:

Security (i)	Portfolio Weight (w_{Pi})	Benchmark Weight (w_{Bi})	ReturnE (R_i)
X	30%	40%	11.20%
y	15%	25%	4.25%
z	55%	35%	14.00%
Total	100%	100%	

- A) 0.44%
 - B) 1.77%
 - C) 1.26%
-

Question #25 of 40

Question ID: 1474056

When choosing an active manager, an investor with a high level of risk aversion:

- A) will choose the manager with the highest information ratio.

B) will choose the manager with the highest active return.

C) will choose a manager with the lowest active risk.

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Question ID: 1474049

Which of the following terms is the number of independent bets per year made by an active manager?

A) Breadth

B) Transfer Coefficient

C) Information Coefficient

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Question ID: 1474075

Pamela Grieve claims that her information coefficient is 0.20 on monthly bets on 10 stocks in the healthcare industry. Assuming unconstrained optimization, the reduction in her information ratio if her bets have a correlation coefficient of 0.45 as opposed to being truly independent is *closest* to:

A) 45%

B) 22%

C) 86%

Question #28 of 40

Question ID: 1474039

Which of the following statements is *least accurate*?

A) Unlike Sharpe ratio, information ratio is affected due to addition of cash or leverage.

B) The information ratio of a constrained active portfolio is unaffected by aggressiveness of the active weights.

Sharpe ratio of a portfolio consisting of a combination of benchmark and actively managed portfolio with positive active return will be higher than the Sharpe ratio of the benchmark.

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Question ID: 1474047

Which of the following terms is the ex-ante risk weighted correlation between forecasted active returns and actual active returns?

- A) Breadth
 - B) Information Coefficient
 - C) Transfer Coefficient
-

Question #30 of 40

Question ID: 1474050

An active manager has an information coefficient of 0.08, transfer coefficient of 0.50, and makes 100 independent bets per year. What is the expected active return for an active risk constraint of 5%?

- A) 1.8%
 - B) 2.0%
 - C) 2.4%
-

Question #31 of 40

Question ID: 1474063

Charles Griffith makes quarterly bets between stocks of industrial and utility sectors. Griffith's strategy has an annualized active risk of 18%. Based on the information below, If Griffith wants to limit his active risk to 6%, what is the allocation to Utility sector when Griffith is bullish about Industrial stocks?

Benchmark	
Sector	Weight
Industrial	80%
Utility	20%

A) -13%
B) 5%
C) 14%

Question #32 of 40

Question ID: 1474048

Which of the following is correct for an unconstrained active portfolio?

- A)** $TC < 1$
B) $TC = 1$
C) $TC > 1$

Sundar Mithai, CFA, is a fund manager for Pearl Investments and makes a monthly report to the firm's partners. Mithai mentions two active managers in his report, Galab and Phasar. Exhibit 1 provides additional information on the two managers:

Exhibit 1: Selected Information on Galab and Phasar

	Galab	Phasar
Information coefficient	0.22	0.37
Transfer coefficient	0.8	0.73

Active risk 5.6% 6.6%

Active return 10.8% 9.2%

Mithai makes the following comments regarding the two active managers:

Comment The investment mandate of Phasar appears to be less constrained relative to
1: Galab.

Comment Galab appears to have better skill at predicting returns.
2:

Comment 1: The investment mandate of Phasar appears to be less constrained relative to Galab.

Comment 2: Galab appears to have better skill at predicting returns.

Mithai recently decided to give all the analysts at the firm a refresher on the fundamental law of active portfolio management. Details of a hypothetical unconstrained fund is shown in Exhibit 2.

Exhibit 2: Hypothetical Fund

Information coefficient 0.14

Monthly active bets 5

Active risk 4.32%

Question #33 - 36 of 40

Question ID: 1474070

According to the fundamental law of active management, how many forecasts is Galab making per month?

- A) 3.
 - B) 10.
 - C) 36.
-

Question #34 - 36 of 40

Question ID: 1474071

How many of Mithai's comments are correct in relation to the comparison between Galab and Phasar?

- A) One.
 - B) Both.
 - C) None.
-

Question #35 - 36 of 40

Question ID: 1474072

The expected active return generated by the hypothetical fund described in Exhibit 2 is:

- A) 3.12%.
 - B) 4.68%.
 - C) 8.20%.
-

Question #36 - 36 of 40

Question ID: 1474073

If the hypothetical fund described in Exhibit 3 was subject to investment constraints, its expected active return would be expected to:

- A) rise.
 - B) fall.
 - C) remain unchanged.
-

Question #37 of 40

Question ID: 1474065

Which of the following statements regarding the information ratio of an unconstrained portfolio is *most likely* correct?

A market timer who uses independent information to make predictions about market movements on a monthly basis and has an information ratio of 0.20 must

A) have an information coefficient higher than a stock selector with the same information ratio who follows 10 stocks and revises his forecast quarterly

A market timer who uses independent information to make predictions about market movements on a monthly basis and has an information ratio of 0.20 must

B) have an information coefficient lower than a stock selector with the same information ratio who follows 10 stocks and revises his forecast quarterly

A market timer who uses independent information to make predictions about market movements on a monthly basis and has an information ratio of 0.20 must

C) have an information coefficient equal to a stock selector with the same information ratio who follows 10 stocks and revises his forecast quarterly

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Question ID: 1474067

Tom Grenkin is a market timer with an information ratio of 0.75. He makes a prediction of the movement in the market each quarter. Jane Fortina is a stock selector who follows 50 companies and revises her assessment each quarter. She also has an information ratio of 0.75. Assuming both managers have unconstrained portfolios, which of the following statements regarding the two managers is *most accurate*?

- A)** As Grenkin makes fewer bets per year, he requires a higher information coefficient on each bet than Fortina to achieve the same information ratio.
- B)** As both managers have the same information ratio, they must also have the same information coefficient.
- C)** As Fortina's strategy has a much larger breadth, she must have a larger information coefficient than Grenkin.
-

Question #39 of 40

Question ID: 1474051

An active manager has an information coefficient of 0.05 and makes 36 independent bets per year. What is the manager's information ratio given a transfer coefficient of 0.75?

- A)** 0.45
- B)** 0.23

C) 1.35

Question #40 of 40

Question ID: 1474053

An active manager has an information coefficient of 0.07, transfer coefficient of 0.90, and makes 49 independent bets per year. Benchmark portfolio has a Sharpe ratio of 0.40 and standard deviation of benchmark returns is 12%. The optimal amount of active risk is *closest* to:

- A) 8%
- B) 6%
- C) 14%